TAO Wenwen

Targeted Academic CV — Neural Decoding BCI and Human AI Interaction

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Research Profile

Clinically trained multimodal neuroscience researcher developing experimental systems for decoding neural, physiological and behavioral states. My work integrates simultaneous EEG–fNIRS, eye tracking, speech, facial video and autonomic sensing with synchronized paradigms, quality control and clinically grounded state labels for future BCI and adaptive human–AI systems.

Research Interests

Neural State Decoding · Multimodal Neuroscience · EEG–fNIRS · Physiological Sensing · BCI · Human–AI Interaction · Adaptive Systems · Neurovascular Coupling

Education

Nanchang University | B.M. in Clinical Medicine | 2017–2022

GPA: 3.2/4.0 (82/100). Relevant training: Psychiatry, Neurology, Internal Medicine, Pathology, Pharmacology, Medical Statistics and Evidence-Based Medicine.

Publications and Manuscripts

1. **Tao, W.** Early Depressive Symptom Severity and Six-Month Return to Drinking After Inpatient Treatment for Alcohol Dependence: A Prospective Cohort Study. Manuscript submitted to *Journal of Substance Use and Addiction Treatment*.
2. **Tao, W.** Temporal Dataset Shift Limits Relapse-Risk Stratification Across the Inpatient-to-Community Transition in Alcohol Dependence: A Temporal Comparison of Conventional Machine Learning and Self-Supervised Representation Learning. Manuscript submitted to *Addiction Science & Clinical Practice*.
3. **Tao, W.** Task state dominates participant-held-out prediction of risky decisions in methamphetamine dependence, with limited incremental value from pre-decision EEG. Manuscript submitted to *Clinical EEG and Neuroscience*.

Research Experience

Tsinghua University — CogDrive Lab | Mar 2026–Present

State Key Laboratory of Intelligent Green Vehicle and Mobility

Research Assistant | Advisor: Prof. Dongpu Cao

Project: *Multimodal AI for Adolescent Depression Screening and Human–AI Interaction in Intelligent Cockpits*

Clinical collaborator: Tongji Hospital, Huazhong University of Science and Technology

- Independently conceived and led an IRB-approved prospective multimodal cohort protocol for children and adolescents aged 6–18, with planned enrollment of 300 clinical participants and 150 healthy controls across four developmental bands; real-world recruitment and acquisition remain in the pre-collection hardware-validation stage.
- Established a four-stage clinical ground-truth framework—CDRS-R, age-matched self/parent reports, K-SADS-PL DSM-5 clinician confirmation and multisource severity grading—and mapped item-level clinical anchors to facial, speech, gaze and physiological features.
- Designed synchronized acquisition across 13+ streams, including simultaneous 64-channel EEG–fNIRS, desktop eye tracking, multi-view facial video, speech, ECG, PPG, GSR and respiration during computerized paradigms and clinician-led interviews.
- Specified event and trigger conventions, tiered temporal alignment, metadata, operator procedures, signal-quality labels and analysis-ready schemas for multimodal state modeling and exploratory EEG–fNIRS neurovascular-coupling analysis.
- Operationalized K-SADS-PL DSM-5 into four age-adaptive structured dialogue protocols spanning seven clinical domains for a planned speech-based multimodal large-language-model workflow.
- Developed BPlus MVP-0 with Rest, MMN, VFT, N-back, Emotion and Recovery modules, participant-response capture, event logging, preflight checks and automated QC/state reports; software demonstrations and dry runs passed, while real-hardware synchronization validation remains pending.
- Designed the intelligent health-cockpit translation architecture linking validated multimodal state estimates to safe in-vehicle interaction, including confidence-aware routing, an L0–L5 safety gate and edge–cloud coordination; product deployment remains contingent on clinical and systems validation.

Shenzhen Mental Health Center — Department of Addiction Medicine | Jan 2024–Aug 2025

Addiction Neuroscience and Translational Research

Research Assistant, Full-time | Advisors: Prof. Mei Yang and Prof. Xiaojian Jia

Project: *Longitudinal Addiction Cohorts, Multimodal Neuroimaging and Gut–Brain Research*

- Managed research workflows for 290+ participants with alcohol use disorder, methamphetamine dependence and matched controls, integrating longitudinal clinical phenotypes, MRI/DTI, EEG, cognition and biospecimens in a standardized database with >95% completeness.

- Investigated early depressive symptoms and six-month return to drinking in 107 inpatients using prospective follow-up and covariate-adjusted models; the work is reported in a manuscript submitted to the Journal of Substance Use and Addiction Treatment.
- Evaluated temporal dataset shift in relapse-risk stratification across inpatient and community phases, comparing conventional machine learning with self-supervised representations and emphasizing transportability rather than in-sample performance.
- Analyzed participant-held-out risky-decision prediction in methamphetamine dependence, finding that task state dominated prediction and that pre-decision EEG provided limited incremental value; the manuscript is submitted to Clinical EEG and Neuroscience.

Tsinghua University — Yan Lab | Sep 2025–Mar 2026

Depression Neuroimaging and Neuromodulation Research

Research Intern | Advisor: Prof. Chaogan Yan

Project: *Mind Flower — rTMS Neuromodulation and Mindfulness-Based Counseling for Depression*

Multi-phase support: NSFC, National Key R&D Program, STI2030 Brain Science and CAS

- Investigated longitudinal clinical and multimodal markers of depression and rTMS treatment response by screening participants and administering MINI, HAMD, HAMA and MADRS assessments at baseline, intervention and follow-up.
- Managed resting-state, rumination and distraction fMRI acquisition, including individualized negative-recall cues, participant-specific E-Prime substitution, real-time head-motion monitoring (<3 mm) and OptiMRI think-aloud audio quality control.
- Tracked repeated rTMS sessions, daily pre-treatment HAMD-6 variation and adverse effects, creating structured dose–response and tolerability records alongside WCST, SST and emotional n-back assessments.

Beijing Normal University — Liu Lab | Sep 2025–Mar 2026

Parkinson's Disease Neuromodulation Research

Visiting Research Intern | Advisor: Dr. Zheyun Liu

Project: *MRI-Guided Transcranial Ultrasound Stimulation of Bilateral GPi for Cognitive Modulation in Parkinson's Disease*

- Supported full-protocol transcranial ultrasound stimulation targeting the bilateral globus pallidus internus using structural MRI and Brainsight neuronavigation for individualized deep-target localization.
- Documented stimulation parameters, including pulse-repetition frequency and duty cycle, and maintained target, session and safety records for reproducible protocol execution.

Clinical Experience

Wannian County Hospital of TCM | Resident Physician | Dec 2022–Dec 2023

- Managed 15–20 chronic-disease inpatients through assessment, treatment planning, inpatient care and follow-up and developed standardized medical-record templates for longitudinal traceability.

Ganzhou People's Hospital | Clinical Intern | Feb 2021–Feb 2022

- Completed 12 clinical rotations, including Neurology ICU and Cardiology ICU, and performed supervised procedures including lumbar puncture and ECG interpretation.

Methods and Technical Expertise

Clinical and Psychiatric Assessment: MINI; K-SADS-PL DSM-5; CDRS-R; HAMD; HAMA; MADRS; PHQ-9; PHQ-15; CDI; SDQ; longitudinal cohort design; IRB coordination; consent/assent; pediatric safeguards

Neuroimaging: Structural MRI; resting-state and task fMRI; DTI tractography; FA/ADC; structural–functional connectivity; FSL; DPABI; DPABI-Surf; MRI-based targeting

Neurophysiology and Multimodal Sensing: 32/64-channel EEG; simultaneous EEG–fNIRS; ECG; PPG; GSR; respiration; desktop eye tracking; synchronized video and speech; event-trigger design; signal QC

Neuromodulation: rTMS; transcranial ultrasound stimulation; Brainsight neuronavigation; individualized targeting; stimulation-parameter logging; tolerability and safety monitoring

Behavioral and Cognitive Paradigms: MMN; VFT; N-back; emotional n-back; WCST; SST; BART; two-step task; E-Prime and PsychoPy workflows

Programming and Quantitative Methods: Python; R; SPSS; MATLAB; logistic regression; longitudinal modeling; participant-held-out validation; temporal transportability; multimodal data schemas

Research Infrastructure and Data Engineering: BPlus MVP-0; synchronization and trigger architecture; metadata and automated QC; longitudinal databases; data dictionaries; version control; NAS archiving; multicenter harmonization

Languages

Chinese: Native | English: IELTS 6.5; TOEFL 97